

Timothy Mai

Swedesboro, NJ | 267-798-9479 | timothy.h.mai@drexel.edu | Handshake | Portfolio

Education

Master's of Science: Data Science
Drexel University, GPA 3.7

Philadelphia, PA
Graduation: June 2027

Bachelor of Science: Computer Science
Rowan University

Glassboro, NJ
Fall 2024

Skills

Languages: Python / Java / C++ / SQL
Web & Frameworks: HTML / Dart / CSS / Dart / Django / Firebase
Data & Visualization: Tableau / Excel / Jupyter Notebook / Microsoft Office Suite
Developer Tools: Git / Github / VS Code

Projects

Weather Impact on Energy Consumption Personal
Software: Python / Pandas / Matplotlib Present

- Developed a **predictive model** to forecast **daily electricity demand** using data from the **U.S. Energy Information Administration (EIA)** and **NOAA weather stations**
- Collected, cleaned, and merged **electricity demand** and **weather datasets** for **time-series analysis**
- Performed **preprocessing**, handled **missing values**, and improved **data quality** for predictive modeling

Portfolio Personal
Software: TypeScript / ASTRO / CSS / JS Present

- Customized an **Astro-based portfolio template** to showcase projects, skills, and experience
- Configured and deployed the site using **Cloudflare Pages** with **Git-based CI/CD**
- Maintained **version control** and updates through **GitHub integration**

Rail Risk Census Analysis Drexel University
Software: Python / Pandas / Matplotlib March 2026

- Integrated **FRA railroad accident data** with **USDA Rural–Urban Continuum Codes** to analyze accident risk across urban and rural environments
- Performed **data cleaning**, **exploratory data analysis**, and **feature engineering** to improve dataset quality
- Prepared structured **train/test datasets** for **predictive modeling** and machine learning analysis

Exoplanet Multimodal Dataset Pipeline Drexel University
Software: Python / Pandas / Jupyter Notebook March 2026

- Developed a **data acquisition and preprocessing pipeline** to integrate **NASA Exoplanet Archive** and **EU Exoplanet datasets** using **TAP-based queries**
- Built **data cleaning** and **merging workflows** in Pandas to construct a **multimodal exoplanet dataset** for scientific analysis
- Organized and processed **large-scale astronomical datasets** in **Jupyter Notebook** to support exploratory analysis and machine learning workflows

Professional Experience

Kung Fu Tea Glassboro, NJ
Manager Jun 2019 - Jan 2026

- Trained and supervised **15+ employees**, coordinating schedules and workflows using **Google Sheets** to optimize team productivity
- Tracked and analyzed **inventory data** to identify trends, streamline ordering, and reduce **stockouts and waste**
- Applied **analytical problem-solving** to resolve operational challenges in a **fast-paced environment**